PUTI WEN

pw1246@nyu.edu | Abu Dhabi, UAE | github.com/putiw | linkedin.com/in/puti-wen

A somewhat unconventional neuroscientist. My PhD was in visual neuroscience, but my recent work has centered on building diagnostic and prognostic reporting tools for MS. I also spend most of my time building processing pipelines and maintaining infrastructure for large neuroimaging datasets. I work across mixed fields: psychology, radiology, engineering, clinical neurology, or whatever comes up.

TECHNICAL SKILLS

Neuroimaging	fMRIPrep, FreeSurfer, TractoFlow, MRIQC, BIDS, dcm2bids, DICOM, HCP, ANTs, Nextflow
MRI Hardware	Siemens scanners, Psychtoolbox, EyeLink eye-tracking, VPixx/ProPixx
Automate Workflow	Docker, Singularity, GitHub Actions
Manage Infrastructure	XNAT, Slurm (HPC), Supabase, Box API
Build Applications	React Native, Godot, Typescript
Analyze & Visualize	Python, MATLAB, Adobe Illustrator, Blender

KEY PROJECTS

XNAT Platform & Brain Imaging Biobank — *NYUAD Center for Brain and Health*

- ***Processing pipelines.*** Built and containerized six processing pipelines (Docker/Singularity) on HPC for functional MRI preprocessing, structural brain segmentation, diffusion tractography, MRI quality control, cortical visual area mapping, and custom data validation with edge-case handling and BIDS compliance.
- ***Automated QA.*** Automated daily workflows for the brain imaging biobank: validation dashboard showing per-subject status updated nightly; automated daily backup of newly processed and validated data from XNAT to BOX.
- ***Cross-site QC.*** Conducted cross-site MRI quality comparisons (Siemens CIMA.X vs NYUAD, UAEU vs NYUAD) evaluating SNR, tSNR, CNR, and GSR metrics. Currently doing QC between CIMA.X and Prisma.
- ***Normative dashboard.*** Preprocessed 9,000+ subjects from HCP and ADNI datasets; built a normative brain dashboard with age/gender-controlled brain structure metrics allowing researchers to compare their data against normative curves.
- ***Documentation.*** Authored technical documentation (350+ commits); personally triage and resolve user-reported GitHub issues over the years.

LesionView & Longitudinal Report — *Clinical MS lesion tracking and automated reporting*

- ***Lesion tracking app.*** Built a clinical imaging application for tracking MS lesions across timepoints; designed and implemented an automated longitudinal lesion change report that identifies new and enlarging lesions with color-coded and animated visualization.
- ***Web migration.*** Migrated from MATLAB prototype to browser-based React Native/Expo app after two external contractors failed to deliver; iterated features weekly with Yas Clinic neurologists.
- ***Patent filing.*** Co-inventor on provisional patent application (NYU Ref: ABD01-03), filed May 2026.

CvsView — *AI-powered central vein sign detection for MS diagnosis*

- ***Diagnostic tool.*** Built a web-based review tool for clinicians to efficiently verify the central vein sign (CVS) in MS lesions, designed to assist current clinical workflow; includes a modular AI component built as plug-and-play for future models.
- ***Data quality & deployment.*** Resolved critical SWI-FLAIR alignment issues by manually re-labeling data after discovering co-registration artifacts; Docker-containerized and deployed as a web service; processes clinical data on demand for Harley Street Medical Centre.

MyPhysio — *Full-stack clinical application for physiotherapy practice management*

- ***Full-stack app.*** Conceived and built a full-stack clinical app (React Native, Expo, Next.js, Supabase) in collaboration with two colleagues; led development; features include admin dashboards, patient portals, exercise prescription, and pain and range of motion tracking.
- ***Deployment.*** Deployed with Hostinger, Supabase, GitHub Actions, Docker.

afqView — *Interactive 3D white matter tractography visualization*

- ***Tractography QC tool.*** A colleague needed to QC tractography data across 100+ subjects, each with 20+ tracts and 10 iterated versions — all reviewed file-by-file. Built a web application with side-by-side before/after tract comparison, per-subject navigation, color-coded availability indicators, and interactive 3D rendering.

Additional Contributions —

- Created 8 multi-material 3D printed brain models (gray/white matter, MS lesions) for patient education; collaborated with NYUAD Engineering on fMRI research; designed CBH poster for NYUAD Research Institute Day, one-pagers for NMSS/LAMINATE stakeholder meetings, CBH promotional video, and MS Research Day event banners, etc.

RESEARCH EXPERIENCE

Postdoctoral Researcher | NYU Abu Dhabi, Center for Brain and Health — Sep 2024 – Present

- Led research on tract-based lesion metrics in multiple sclerosis, combining quantitative MRI with machine learning to predict clinical disability; co-authored 4 MS neuroimaging papers
- Provided technical training to 10+ researchers across BIDS, fMRIPrep, FreeSurfer, XNAT, and HPC workflows; conducted formal workshops for research assistants on data validation procedures
- Configured MRI stimulus computer hardware, resolving Psychtoolbox compatibility on Apple Silicon; built template experiments with EyeLink eye-tracking, VPixx button box, and ProPixx projector integration
- Co-investigator on DOH grant for optic nerve myelination characterization (collaborative with CCAD and Polytechnique Montreal, submitted Jan 2026)
- **Science communication:** Invited talk at WHX Arab Health 2026 Total Radiology Conference; panelist on "AI in Neuroradiology"; presented to Siemens Healthineers delegation and NYUAD provost; represented ITH startup at GITEX YouthX and Hub71; lecturer at CBH Brain Health Workshop (fMRI module), MS Workshop, and Brain Imaging Workshop; posters/talks at ISMRM 2025 (Honolulu), OHBM 2025 (Brisbane), MENACTRIMS 2025 (Dubai), Innovations in Neuroimaging 2026; 1st Place poster at Division of Science Annual Conference 2026

Ph.D. Researcher | New York University, Dept. of Psychology — Aug 2019 – Aug 2024

- Investigated neural pathways of 3D motion perception in humans using fMRI; developed novel single-trial decoding methods and multivariate pattern analysis pipelines
- Published 3 first-author papers in top neuroscience journals (J. Neuroscience, Imaging Neuroscience, NeuroImage)
- Built motion localizer stimuli and analysis pipelines (Psychtoolbox/MATLAB) adopted across the lab
- **Science communication:** Presented at Vision Sciences Society (VSS) three consecutive years (2022 talk, 2023 and 2024 posters); poster at MENACTRIMS 2024 (Jeddah); lecturer at Brain Imaging Workshop (2023); TA for Lab in Visual Neuroscience; Invited speaker for advanced MRI methods in MS (2024).
- Advisors: Bas Rokers, Michael S. Landy

Research Assistant | Vanderbilt University, Randolph Blake Lab — 2016 – 2018

- Conducted research on multisensory perception in the lab of Randolph Blake, investigating temporal judgments of audiovisual events; first-author publication in Vision (2020)

EDUCATION

Ph.D., Cognition and Perception | New York University — 2019 – 2024

B.A., Psychology (Major), Mathematics (Minor) | Belmont University — 2015 – 2018

Computational Neuroscience: Vision Workshop | Cold Spring Harbor Laboratory — 2022

INTELLECTUAL PROPERTY & GRANTS

- **Co-Inventor, Provisional Patent Application:** LesionView (NYU Ref: ABD01-03) and Automated Cloud-Based MS MRI Pipeline / LAMINATE (ABD01-04). Filed May 2026 via NYU Technology Opportunities & Ventures.
- **Co-Investigator, DOH Grant:** "Diagnosis of optic neuritis through biophysically-informed characterization of optic nerve myelination." Collaborative with CCAD and Polytechnique Montreal. Submitted Jan 2026.

SELECTED PUBLICATIONS

5 peer-reviewed journal articles (4 as first author), plus 10+ total published conference abstracts and preprints (bioRxiv, medRxiv).

Wen, P., Thompson, L. W., Rosenberg, A., Landy, M. S., & Rokers, B. (2025). Single-trial fMRI decoding of 3D motion with stereoscopic and perspective cues. *Journal of Neuroscience*.

Wen, P., Ezzo, R., Thompson, L. W., Rosenberg, A., Landy, M. S., & Rokers, B. (2025). Functional localization of visual motion area FST in humans. *Imaging Neuroscience*.

Wen, P., Landy, M. S., & Rokers, B. (2023). Identifying cortical areas that underlie the transformation from 2D retinal to 3D head-centric motion signals. *NeuroImage*.

Wen, P., Opoku-Baah, C., Park, M., & Blake, R. (2020). Judging relative onsets and offsets of audiovisual events. *Vision*.

Mohamed, A. Z., ..., Wen, P., et al. (2026). The ASPIRE Research Institute Dataset: Building a foundation for brain health research in the UAE. *Scientific Data*.

Abdullah, O.*, Wen, P.*, et al. (2026). Tract-based lesion metrics for resolving the clinico-radiological paradox in multiple sclerosis. *bioRxiv*.

PERSONAL

Languages: English, Mandarin Chinese, basic Turkish. **Design:** T-shirt design selected for CSHL Computational Neuroscience Workshop (2022); designed CBH t-shirt/hoodie (2025); artwork featured in fashion shows in China. **Other:** Science YouTube channel (N.A.R. Things); built a self-sustaining aquarium requiring no filter or water changes; former top-ranked World of Warcraft player in the US.